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Aerothermodynamics of a Low Earth Orbiter Launcher: Preliminary Heating Analysis

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Abstract

The report investigates the preliminary heating analysis and subsequent design of a Thermal Protection System (TPS) for a low Earth orbit launcher. Using a method known as the Centreline Heating methodology, heating is analysed on ascent and first stage EDL, following separation at 79 km. The peak temperatures and heat flux during descent were found to be 1670 K and 44 W/cm² respectively. On ascent, a peak temperature of 1200 K was observed. The results compare favourably to CFD, to within 15 %. A heat shield for the first stage was designed from LI-900, weighing 30 kg. A feasibility study into recovery of the second stage found it uneconomical due to the excessive weight of the heat shield required. Acknowledgement is given to the approximate nature of the estimations, with CBAERO or FIAT being identified as computer software able to better predict thermal loads.

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Nomenclature

The following list describes several symbols that will be later used within the body of the document. Units are as given unless stated otherwise.

Symbols

A	Cross-sectional area	m^2
BC	Ballistic Coefficient	kg/m^2
C_D	Drag coefficient	
c_P	Specific heat capacity	$J/kg \cdot K$
C_P	Pressure coefficient	
$C_{P_{max}}$	Post-shock pressure coefficient	
h	Specific enthalpy	J/kg
k	Thermal conductivity	$W/m \cdot K$
m	Mass	kg
M	Mach number	
Pr	Prandtl number	
P	Pressure	Pa
$\dot{q}$	Rate of heat transfer per unit area	W/m^2
r	recovery factor	
R_c	Radius of curvature	m
Re	Reynolds number	
s	Wall tangential coordinate	m
t	Time	s
T	Temperature	K
V	Flow velocity	m/s
α	Thermal diffusivity	m^2/s
ε	Emissivity	
γ	Ratio of specific heat capacities	
ϕ	Flow deflection angle	rad
θ	Momentum boundary layer thickness	m
μ	Dynamic viscosity	$Pa \cdot s$
ρ	Density	kg/m^3

Sub-indices

a	Adiabatic conditions
e	Edge of boundary layer conditions
T	Transition
w	Wall conditions
0	Stagnation conditions
∞	Free stream conditions

Super-indices

$\star$	Conditions at reference temperature
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Constants

σ	Stephan-Boltzmann constant	$5.670367 \times 10^{-8} W/m^2 K^4$
C	Radiative heat transfer constant	4.736×10^{-8}

Abbreviations

CFD	Computational Fluid Dynamics
DoF	Degree of Freedom
EDL	Entry, Descent and Landing
TPS	Thermal Protection system

1 Introduction

Recently there has been an increased drive towards reusability of spacecraft, in an attempt to reduce the amount of space debris and drive down costs, with incentives being given by the EU [1] for missions which fulfil their criteria. The PICASSO project aims to explore the growing market for micro-satellites, or CubeSats, and aims to have its first launch in 2022.

During use, a spacecraft is subjected to considerable heating which must be accounted for in the initial design stages to prevent failure of the structure. This report is primarily concerned with the estimation of this heating and subsequent design of a heat shield which is able to withstand and protect the spacecraft from thermal loads during EDL.

One of the main aims of this project is to maximise reusability of the spacecraft. In this case it was decided that only the first stage would be recovered as this had the strongest business case [2]. Thus this report primarily focuses on the initial heat shield design of the first stage, with a feasibility study being carried out into the second stage. This report should be read alongside those by Gomez [3] and Kent [4] which focus on the hypersonic and subsonic aerodynamic analyses, stability and CFD simulations.

Objectives

The following objectives are identified for successful prediction of heating and design of the heat shield:

- Calculation of the descent trajectory of the first stage
- Estimation of the heating on the first stage during EDL
- Calculation of the maximum temperatures the heat shield would be exposed to
- Design of a heat shield able prevent the spacecraft structure from reaching extreme temperatures
- Analysis of the heating on the ascent trajectory
- Investigation into the feasibility of bringing back the second stage

2 Trajectory Analysis

To analyse the heating over a body, its descent trajectory must first be known. The return trajectory of the first stage is calculated using a 3 degree of freedom model [5] which takes into account a spherical, rotating earth.

Ballistic re-entry

The re-entry is modelled as a ballistic (non-lifting) body for which a critical parameter identified in this model is the Ballistic Coefficient (BC), defined in [6] as in Eq. (1). The BC effectively quantifies the the magnitude of aerodynamic effects on the body. A larger BC suggests the body is less effected by aerodynamic drag.

$$BC = \frac{m}{C_d \cdot A} \quad (1)$$

The EDL trajectory of the first stage is based of a first stage separation altitude and velocity of 79 km and 3883 m/s respectively provided by the systems team [7]. Thus plots of Mach number vs altitude can be produced for varying BC, as seen in Figure 1 where the Mach number at a given altitude is found using the ISA model [8].

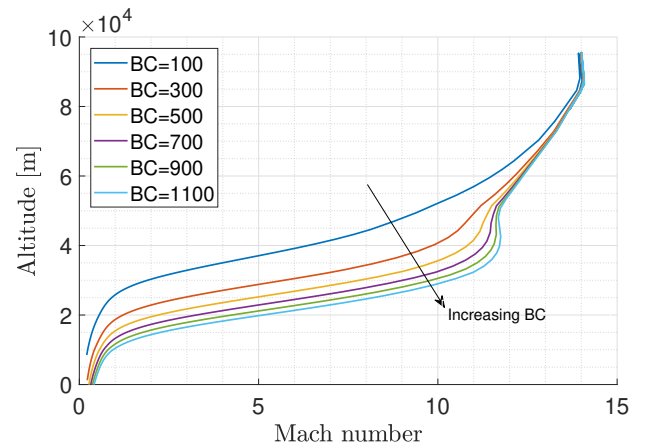


Figure 1: EDL trajectory of first stage for varying BC

Dynamic pressure loads

Clearly bodies with a higher BC are travelling faster lower in the atmosphere. As such, one might expect higher dynamic pressure loads and higher heating, as density is greater in the lower atmosphere. This is shown in Figure 2 for which a higher BC leads to a higher peak dynamic pressure.

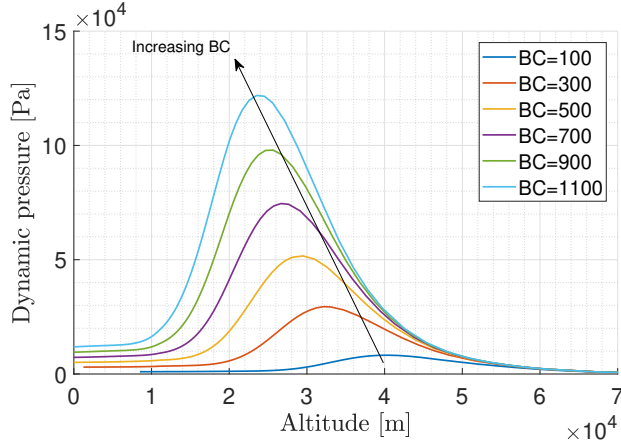


Figure 2: Variation in Dynamic pressure along trajectory for first stage for varying BC

Deceleration loads

One last comment to make on the EDL trajectory is that the body is subjected to high rates of deceleration, leading to high G loads on descent, and is a function of BC as shown in Figure 3. This is not a huge problem in this case since the first stage is returning empty without either payload or crew, thus the only constraint would be structural integrity.

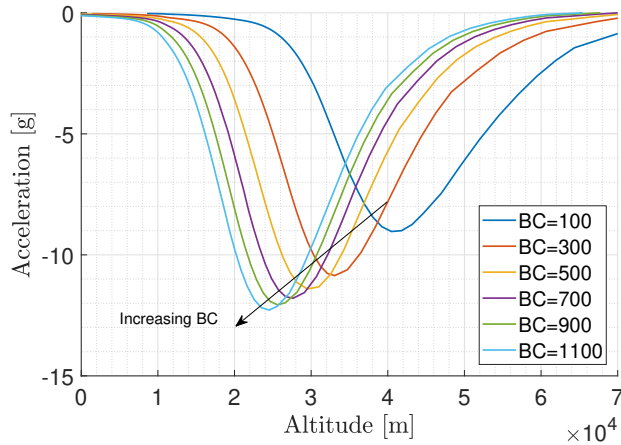


Figure 3: Variation in deceleration along trajectory for first stage for varying BC

3 Final design

For the following section the heating analysis is carried out on the finalised re-entry geometry as shown in Figure 20 Appendix A after first stage separation. The re-entry body geometry is hemispherical, as this avoids having any sharp corners which would lead to increased heat flux

as discussed further in [3].

The first stage empty mass is 986 kg with a diameter of 1.2 m. For the first stage the C_d is calculated to be 0.9 [3]. This means the BC of the empty first stage is 958 kg/m². The final trajectory of the first stage after separation is shown in Figure 4, where peak dynamic forces and G-forces have also been identified.

Table 1: Summary of EDL parameters

Separation altitude [km]	Separation velocity [m/s]	BC [kg/m ²]
79	3883	958

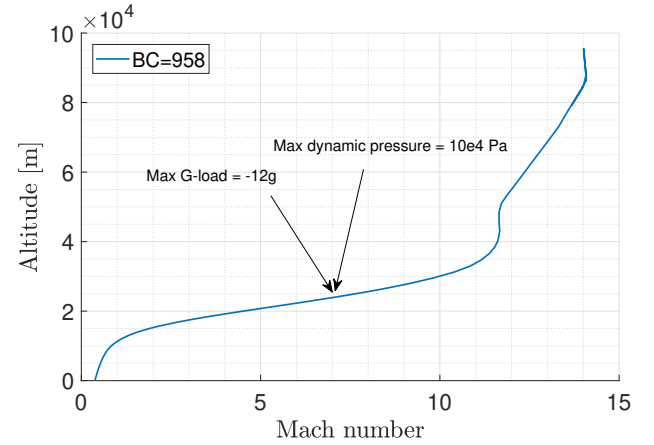


Figure 4: Finalised return trajectory of first stage

4 Heating estimation

At hypersonic speeds, the rate of heat transfer from the air to a body is highly dependant on the boundary layer thickness and conditions along the edge of the boundary layer. In particular pressure, density, velocity and Mach number should all be found at the edge of the boundary layer. Throughout this analysis c_p and γ are taken as a function of temperature [9].

Pressure distribution

In order to find the pressure distribution over a body modified Newtonian theory is used as described in [10] for which

$$C_p = C_{p_{max}} \sin^2(\phi) \quad (2)$$

where $C_{p_{max}}$ is given as a function of free stream Mach number. Using Eq. (2) the boundary layer

edge conditions can then be found using equations 19-24 of [11].

Centreline Heating methodology

In conjunction with [3, 4] a method known as the Centreline Heating method is used to predict heat transfer or heat flux over a generic body. The heat flux over a body is found to depend on what regime the flow is in. In this case four different flow regimes were identified:

1. Stagnation point conditions
2. Near stagnation point conditions
3. Laminar flow
4. Turbulent flow

Stagnation point heat flux

Two methods were identified to evaluate the stagnation point heat flux. The first is a simple method suggested by Sutton & Graves [12] for which the heat flux is given by Eq. (3)

$$\dot{q}_w = k \cdot \left(\frac{\rho_\infty}{R_c} \right)^{1/2} \cdot V_\infty^3 \quad (3)$$

It is worth noting that the power of the free stream velocity V_∞ is often debated [13], but in all cases remains at a value of or just above 3. Another possible scheme is that mentioned by Fay & Riddell [14] in which the stagnation and just off stagnation point heat flux depends on boundary layer edge conditions, free-stream conditions and temperature gradients. The heat flux is then defined as in Eq. (4)

$$\dot{q} = \frac{0.763}{Pr^{0.6}} (h_{aw} - h_w) \sqrt{\frac{\rho_e \mu_e}{R_c} \left(\frac{2(P_e - P_\infty)}{\rho_e} \right)^{0.5}} \quad (4)$$

Emphasis should be put on the fact that for both schemes the heat flux is proportional to one over the square root of the radius of curvature R_c . This explains why re-entry bodies are typically very blunt, as they have very high radii of curvature thus minimise heating. In this analysis Eq. (4) is favoured over Eq. (3) as it is valid for both regimes 1 and 2. A comparison of the schemes can be found in [4].

One last comment to make is that the heat flux is dependent on the temperature gradient

as $h = C_p T$ hence $(h_{aw} - h_w) = C_p(T_{aw} - T_w)$. The adiabatic wall temperature is found using a recovery factor [15] r , relating T_e , T_0 and T_{aw} . For laminar boundary layers it is given by $r = \sqrt{Pr}$ where the Prandtl number can be taken to be constant and equal to 0.71 for hypersonic flows [16].

Off stagnation point heat flux

As mentioned, Eq. (4) is only valid for near stagnation conditions (unless the body is a sphere, for which the solution is exact). This requires a change in the way heat flux is calculated away from the stagnation point on a generic body. For this, the heat flux is estimated in the laminar regime as by Zoby, Moss & Sutton [17] using Eq. (5).

$$\dot{q}_L = 0.22(Re_{\theta_e})^{-1}(\rho^*/\rho)(\mu^*/\mu)\rho_e V_e \times (h_{aw} - h_w)(Pr_w)^{-0.6} \quad (5)$$

for which the Reynolds number is based on the momentum boundary layer thickness defined as in Eq. (6)

$$\theta_L = 0.664 \frac{(\int_0^s \rho^* \mu^* r^2 ds)^{1/2}}{(\rho_e V_e r)} \quad (6)$$

These relations are derived from a modified Reynolds analogy form, in which the heating coefficient is obtained from the skin friction coefficient based on Re_θ . Furthermore, note that in Eqs. (5,6) the heat flux is dependent on variables labelled with an asterisk. This is because they are calculated using Eckert's reference temperature [18] and is further explained in [3]. This is because at hypersonic speeds there are a very large temperature gradients present across the boundary layer, and as such the question becomes for what conditions should the flow properties be calculated, as the heat flux is seen to be highly dependent on local flow conditions. The reference temperature provides a simple analytical solution and is seen to provide reasonably accurate estimations of heat flux.

Transition criteria

It is important to check if the flow has transitioned to turbulence, as turbulent flows lead to enhanced mixing and increased heat flux. An alternative explanation comes from the fact that turbulent flows lead to a higher skin friction coefficient thus via the Reynolds analogy also increases the heat transfer coefficient. The transition criteria used in this report is based on a study of cones in hypersonic flows [19]. The flow is said to have transitioned when the Reynolds number at any point on the body exceeds the transition Reynolds number in Eq. (7)

$$\log_{10}(Re_T) = 6.421 \exp[1.209 \times 10^{-4} M_e^{2.641}] \quad (7)$$

For the re-entry body the boundary layer edge Mach number can be found as previously [11]. Knowing the return to Earth trajectory the free stream Mach number is found for every point in time and so M_e can be calculated. Only the Reynolds number at the point furthest away from the stagnation point is checked, as if this point hasn't transitioned then neither will the rest of the body. This allows transition Reynolds number to be plotted against true Reynolds number along the trajectory to check for turbulence as in Figure 5.

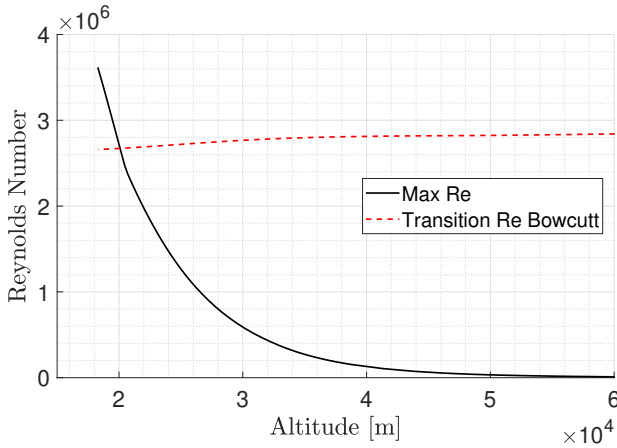


Figure 5: Check for transition during descent

Clearly transition does not occur until very low altitudes for which the stage is at low Mach numbers where the prior analysis is not valid anyway. An alternative transition criteria is that used on the space shuttle [16]:

$$\frac{Re_{\theta T}}{M_e} = 100 \quad (8)$$

Eq. (5) is used over Eq. (8) as it allows calculation of the transition Reynolds number directly without involving the momentum boundary layer thickness and hence enables easier identification of *where* along the body transition will occur.

As is explained later, the Thermal Protection System (TPS) is designed based on heat flux and heat load found for the hypersonic region of EDL. The heating analysis is only carried out for hypersonic speeds (for which modified Newtonian is valid). From trajectory analysis, the body is found to be below hypersonic speeds by around 25 km at which point the flow has not transitioned and as such the turbulent heat flux equations are omitted in this report.

Iteration Procedure

It is now possible to predict the heat flux over the entire body. However, note that in all regimes the heat flux is dependant on the temperature gradient across the boundary layer, requiring knowledge of the surface wall temperature. This is initially assumed to be equal to the free-stream temperature. For applications such as re-entry, the surface temperature is expected to vary along the descent trajectory, and as such the true wall temperature is calculated according to radiative equilibrium [11] where:

$$T_w = \left(\frac{\dot{q}}{\varepsilon \sigma} \right)^{0.25} \quad (9)$$

With the new wall temperature a new heat flux can be found and this iteration is carried out until the wall temperature converges to a final value.

Radiative heating

The prior heating analysis has assumed all heat flux occurs via convective transfer through the boundary layer. It is also possible that some heat flux occurs via radiative heat transfer for which the stagnation point heat flux can be estimated as by Tauber [20] as in Eq. (10)

$$\dot{q} = C R_c^{0.6} \rho^{1.2} f(V) \quad (10)$$

In which $f(V)$ has been tabulated [20]. Figure 6 shows the expected radiative heat flux at an altitude of 79 km (separation altitude) for varying velocities.

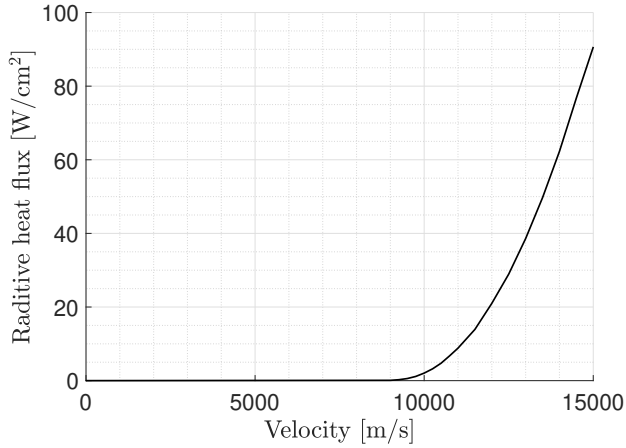


Figure 6: Effect of velocity on radiative heat flux

Clearly below 9000 m/s there is very little radiative heat flux from the air to the body. Given that at separation the first stage is travelling at just 3800 m/s the contribution from radiative heat transfer is neglected in the analysis.

5 TPS design

Having now developed a method to find the heat flux over a body for given free-stream conditions, it is now possible to design a heat shield which would protect the spacecraft structure from the extreme temperatures encountered during descent. Before doing so, a review into existing technologies is carried out after which a selection is made based of design criteria specified later.

EDL strategy

For this design, it was decided to bring the first stage back nose-first. This is an unconventional method, especially for such a long body as it results in a highly unstable EDL, as discussed in [3]. However, in this case it was deemed the best choice as reusability was a key design driver. If the stage was brought back engines first the nozzles would suffer from high heating and become damaged. It is very difficult to design a heat shield that would protect the nozzles, and the geometry of the nozzles is such that there are many sharp edges, which is known to cause concentrated regions of high temperature flow. By coming in nose-first a heat shield must be designed for the first-stage. This will sit directly on top of the fuel tank as they have the same geometry (hemispheric) and thus save space. A

number of options are available for the type of heat shield used and are subsequently discussed and compared.

Deployable vs non-deployable

Recently there has been increased interest towards the use of deployable heat shields [21, 22], the logic being that for a given design the EDL mass is fixed, and C_d is approximately constant. Thus the only parameter the designer has to play with to vary BC is the cross-sectional area, which can be achieved via use of deployable heat shields.

There are numerous benefits to using deployable heat shields during re-entry. The heat shields can be configured to provide some form of lifting EDL, rather than ballistic EDL, decreasing peak G-loads and increasing controllability on landing. Furthermore, it decreases the peak temperatures reached. It may do so enough to prevent/reduce ionisation of surrounding air and facilitate communications and GPS-based navigation systems for longer periods of the descent trajectory [21]. There are two main concepts for deployable heat shields; mechanically deployable heat shields and inflatable aero-shells.

Both methods are valid approaches and have been validated in small scale tests [23, 24] however, both will require additional weight in terms of support infrastructure and mechanisms and thus should only be used if heating is found to be excessive. For the geometry and trajectory being considered the peak heat flux is found to be manageable even if the heat shield is just the diameter of the first stage. Thus, the rest of this report focuses on the design of a rigid, non-deployable heat shield.

Ablatives vs Thermal sinks

There are two material classes when looking into heat shield design. One option is to use so called ablative materials. These are materials which dissipate energy by pyrolysis, an irreversible process in which the material changes state from solid to gas and removes energy from the surface in the process. During this process a gaseous by-product is formed which creates a cooler boundary layer, moving the hot free stream gas away from the surface and reducing heat

transfer. This process is collectively known as blockage. Further cooling is achieved as the TPS materials surface chars while the bulk pyrolyses. A diagrammatic explanation of this process can be seen in Figure 7 [25].

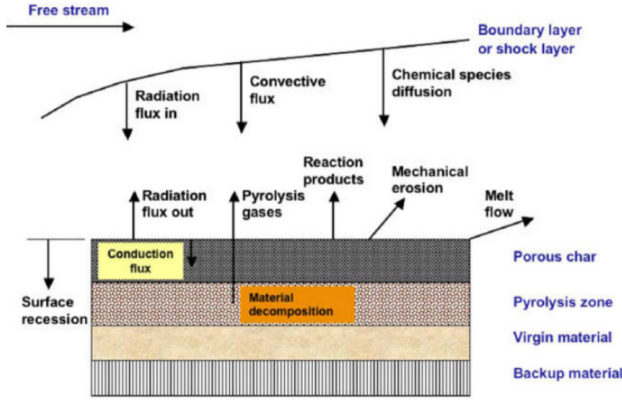


Figure 7: Summation of ablative TPS

Of particular note is that in order for ablatives to work effectively, they must be exposed to heat fluxes high enough such as to cause pyrolysis. If pyrolysis does not occur then care must be taken to ensure that the bondline temperature (temperature at the interface between the TPS and structure of the rocket) does not exceed a critical temperature for a given material and cause failure.

The other option is to use thermal sinks. These are materials which absorb heat and re-radiate it outwards to maintain low surface temperatures. They tend to have very low thermal conductivities, which means that even if exposed to very high temperatures on one side, temperatures are kept comparatively low on the other. Often these are ejected in the last stages of flight to prevent the internal temperatures of the spacecraft rising any further. However, in this case ideally the heat shield should remain attached to maximise reusability.

Comparison of materials

Three materials are initially considered for heat shield selection: Phenolic Impregnated Carbon Ablative (PICA), LI-900 and Acusil II. Both PICA and Acusil II are ablative materials whereas LI-900 is a thermal sink. The properties of all three materials are summarised in Table 2 [26]. All three materials have already been shown

to be successful heat shields, with LI-900 being used on the space shuttle during re-entry [27], Accusil II on the forebody of all mars probes [26] and more recently a variant of PICA being used on Space-X's dragon [28].

Table 2: Summary of material properties

Property	PICA	LI-900	Acusil II
Density [kg/m ³]	274	144	256
Max heat flux [W/cm ²]	1200	75	100
Min Thickness [cm]	3.27	0.686	0.614

Design criteria and heating results

When designing a heat shield there are two parameters which are used in initial sizing. The peak heat flux determines which material is suitable for the expected heating on the spacecraft. The heat load is the integral of the heat flux on the body over time, i.e. a measure of the total "heat" the body is exposed to and determines the thickness of material required. A summary of the heating parameters found for first stage EDL is given in Table 3.

Table 3: First stage EDL stagnation point heating results

Max temperature [K]	Max heat flux [W/cm ²]	Heat load [J]
1670	44	2×10^7

Further discussion of heat load can be found in [4]. Figure 8 shows the calculated variation in stagnation point heating along the descent trajectory for the blunt body shown in Appendix A. Of particular note is that the heat flux initially increases as the first stage descends and enters the denser lower Earth atmosphere but then decreases again as it slows down. Furthermore the stagnation heat flux is only plotted down to 20 km as below this the body is no longer travelling at hypersonic speeds, and the Newtonian pressure distribution is no longer valid. This is not really

an issue though as by this point the heat flux can be seen to be decreasing and is negligible in comparison to peak heat flux experienced.

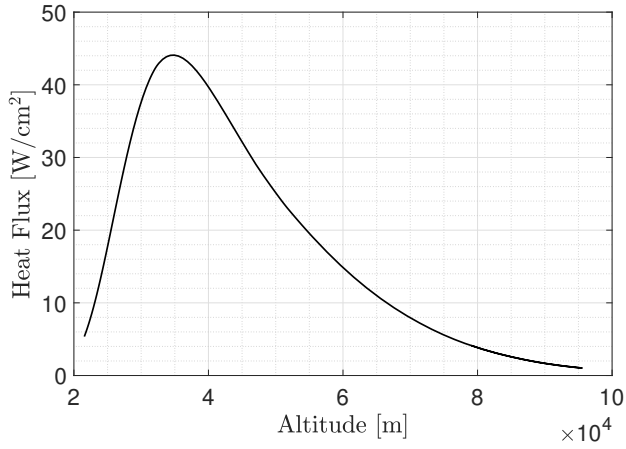


Figure 8: Stagnation Point heat flux along descent trajectory

Since the heat shield parameters are expected to be functions of the heat flux along the body, it is desirable to plot heat flux against position along the nose. Using the Centreline Heating methodology outlined previously, Figure 9 shows the distribution of heat flux along the nose for the point of maximum stagnation heat flux, corresponding to an altitude of 35 km.

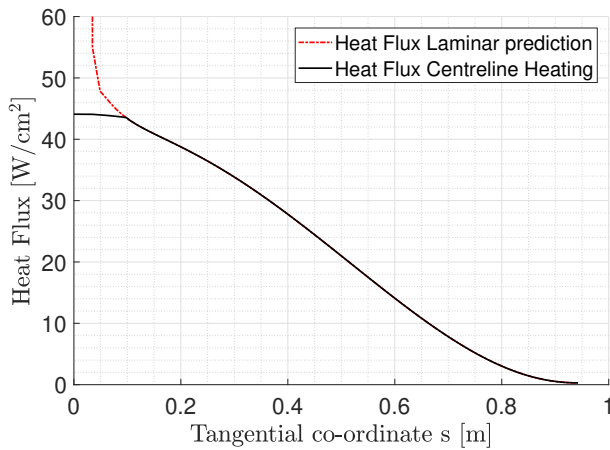


Figure 9: Heating distribution over the body

This figure nicely illustrates the need to use different heating estimation methods for different points along the body. Due to the dependence on Re_θ^{-1} , the equation used in the laminar regime predicts infinite heating at the nose. Clearly this is not the case hence why the method used by Fay & Ridell [14] is used at near stagnation point

conditions instead. The change between regimes is at $s \approx 1$ cm. The criteria for when to change from one regime to the other is simply when the heat flux predicted by both regimes is equal.

As discussed, the heat load over the body is used in sizing the heat shield. Since the heat load is dependant on the heat flux history at any given point, it should be calculated for each point along the nose. For first stage EDL the stagnation point heat flux is found to asymptote to roughly 2×10^7 J/m². The variation in heat flux along the EDL trajectory can be seen in Appendix B. The asymptotic behaviour is explained by looking at Figure 8 at which for an altitude of 21 km heat flux is minimal and so heat load should be expected to remain more or less constant below this altitude.

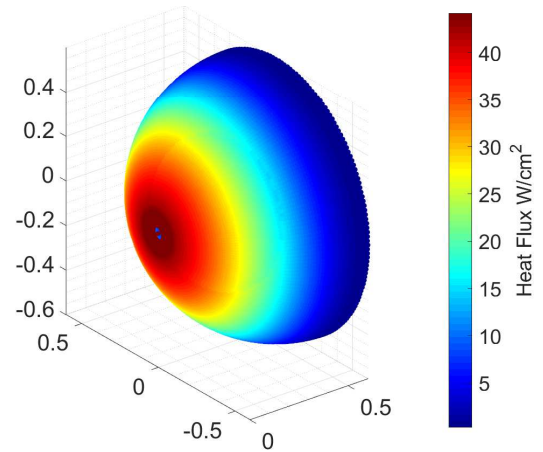


Figure 10: Distribution of heat flux over 3D body

Figure 10 shows the distribution of heat flux around the re-entry body at the point of peak stagnation heat flux. This plot validates theory in that peak heating is expected around the tip of the nose and as such the heat shield is expected to be thickest here.

The temperature profile of the first stage EDL can be found in Figure 23 Appendix B. The peak temperature the heat shield would be exposed to was calculated to be 1680 K. The stagnation temperatures initially decreases after separation as the spacecraft still has some vertical velocity and starts slowing down. As it starts to turn back to Earth it picks up speed and hence has increasing stagnation temperatures.

Heat shield design and material selection

The empirical relationships given in [26] can be used as an initial estimate for the heat shield thickness. Figure 11 shows the distribution of the heat shield required for the three different materials considered.

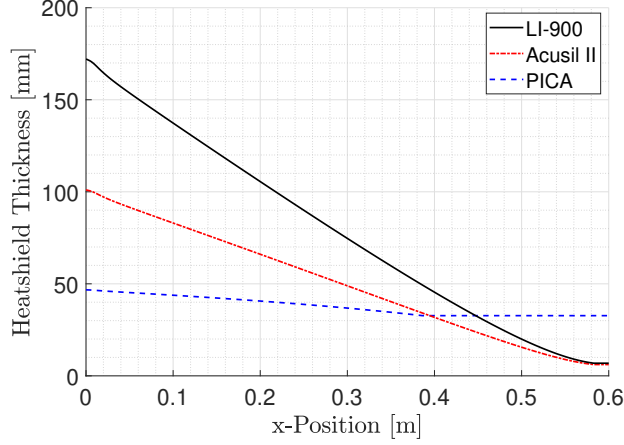


Figure 11: Thickness distribution of heat shield along the body for different materials

With these distributions, the total volume of material required can be found and thus an estimate for the mass of the heat shield can be obtained. Using the densities stated in Table 2, the masses in Table 4 are found.

Table 4: Mass of heat shield for different materials

Material	PICA	LI-900	Accusil II
Mass [kg]	24.9	30.5	31.2

Despite the LI-900 requiring a greater thickness distribution as shown in Figure 11 it is not the heaviest option as its density is half that of the other two options. There were a number of considerations when selecting a final material.

Firstly, one of the main goals is reusability of the first stage. If an ablative material is used the heat shield is lost every time and requires replacement, however a thermal heat sink such as LI-900 can be reused, provided it is not jettisoned or damaged upon return.

Secondly when using ablatives recession must be modelled (the rate at which material is lost)

to see what impact if any this has on the design, which causes added complexity. Lastly ablative materials primarily use pyrolysis as an energy dissipation method. Hence, if the temperatures reached during descent are not high enough for ablation, there is a risk that the material may exceed its bondline temperature.

A common value quoted for when PICA begins to ablate is 2200 °K [29]. The maximum wall temperature reached for the first stage is only 1680 K. Clearly there is a risk of failure if an ablative is used. For the three reasons mentioned it was decided to use LI-900 as the primary heat shield material.

Using the empirical relations in [26] the final geometry of the heat shield is found and shown schematically in Figure 12, for which the heat shield varies from 17 cm thick at the stagnation point to under 1 cm thick at the edges. At first glance this may seem excessive but for comparison the space shuttle had LI-900 tiles upto 12 cm thick [30]. Considering the peak temperature in this case is 1680 K, 17 cm seems reasonable if not a conservative first estimate for the thickness. Further explanation of the seemingly large value can be found in the discussion.

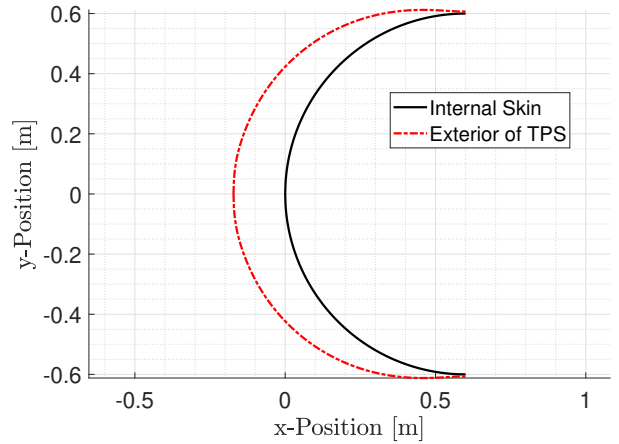


Figure 12: Schematic of heat shield layout

Evaluation of heat shield effectiveness

In this section, analysis is carried out at the stagnation point to measure the effectiveness of the heat shield, as this is where temperatures are highest and so is the critical case. To model the temperature distribution through the heat shield the one dimensional diffusion equation, Eq. (11),

was used.

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad \alpha = \frac{k}{\rho c_p} \quad (11)$$

The thermal diffusivity of LI-900 was calculated using the values in Table 5 [31].

Table 5: Thermal properties of LI-900

Property	k [W/mK]	ρ [kg/m ³]	c_p [kJ/kg·K]
Value	0.05	144	0.9

An initial interior temperature of 200 K, the atmospheric temperature at 79 km [8], is assumed for the rocket internal temperature. In order to estimate heating for the full trajectory, a linear decrease in stagnation wall temperatures is assumed for the non-hypersonic regime as the prior heating analysis is not valid at this point. Using an explicit finite difference scheme the through thickness temperature variation of the heat shield is found as in Figure 13.

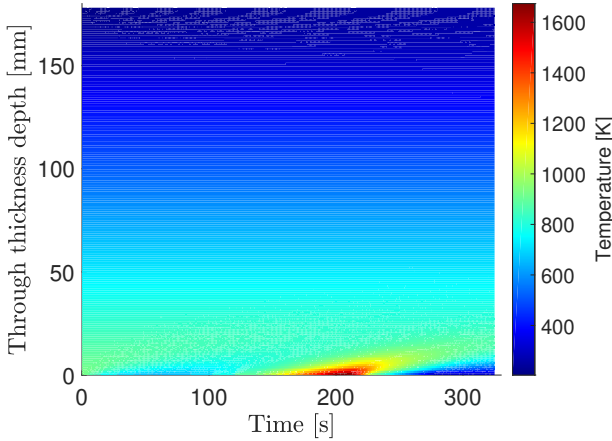


Figure 13: Through thickness temperature variation with time

This shows that the external temperatures hardly penetrate the heat shield at all. This is not surprising given that the body is exposed to heating for less than six minutes, so there should be very little, if any, increase in internal wall temperatures. Internal heating may be more of an issue if a longer descent trajectory is considered or the mission involves sustained high speed flight, e.g skipping re-entry or hypersonic waveriders. The variation of internal body

temperatures during descent is plotted in Figure 14. An increase of just 50 K is found during decent, suggesting the heat shield will be more than enough to protect internal structures.

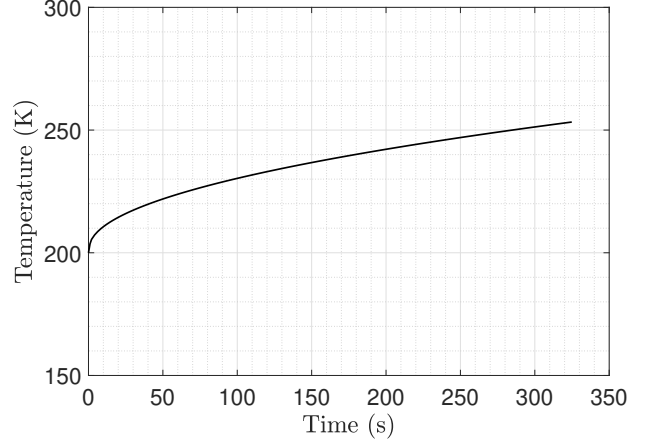


Figure 14: Variation of internal wall temperatures with time

6 Ascent Analysis

Until this point only heating on descent has been analysed, however the spacecraft reaches considerable speeds on ascent as well, and its worth checking if heating will be a problem. For this the ascent trajectory analysed is shown in Figure 15 [7].

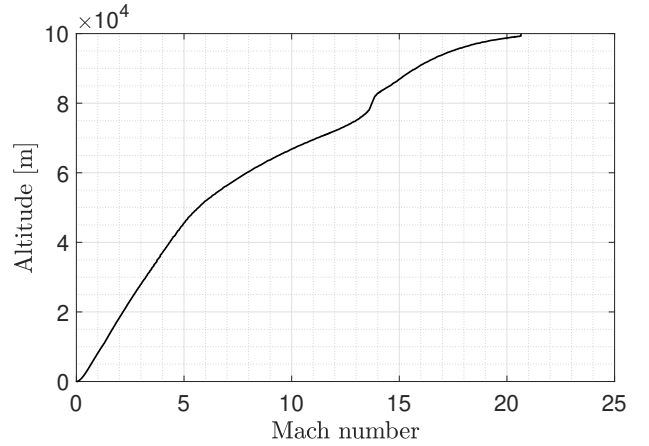


Figure 15: Ascent trajectory

Stagnation point temperatures

Using the same Centreline Heating method as before, the stagnation temperatures that the nose cone would be subjected to on ascent are be calculated. Notice that on ascent the spacecraft had a sharper LD-Haack nose as

detailed in Figure 21 Appendix A. Figure 16 shows stagnation temperature profile for ascent. This is only calculated for hypersonic speeds for reasons explained previously.

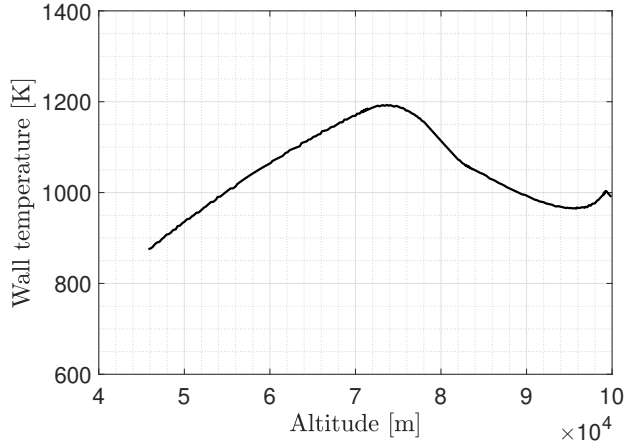


Figure 16: Stagnation temperatures during ascent

Nose cone design

From Figure 16, the spacecraft is found to reach a peak of 1200 K at an altitude of 75 km. It is worth noting that the rocket is only exposed to these temperatures very briefly though and as such a heat shield is unlikely to be needed during ascent.

This is further evidenced when looking at Figure 17. Due to the sharp nose cone on ascent (hence low radius of curvature), heat flux is found to be relatively high near the stagnation point, but very quickly decreases to acceptable levels further away. This results in a similar temperature profile, in which the peak temperature of 1200 K quickly drops to 800 K just 10 cm away from the stagnation point and to atmospheric conditions at the end of the nose. The material chosen for the nose fairing is Carbon fibre reinforced Carbon matrix composite as chosen by the structures team [32] with a maximum operating temperature of approximately 2300 K [33]. In this case the heating during ascent is found to be inconsequential.

A possible problem with a ceramic fairing is the brittle nature of the material. Given that only the very tip of the fairing is exposed to high temperatures, it is possible to make the first 10 cm of the nose from a ceramic, and the rest out of a more conventional material, though there may be issues with manufacturing and bonding

between the two materials.

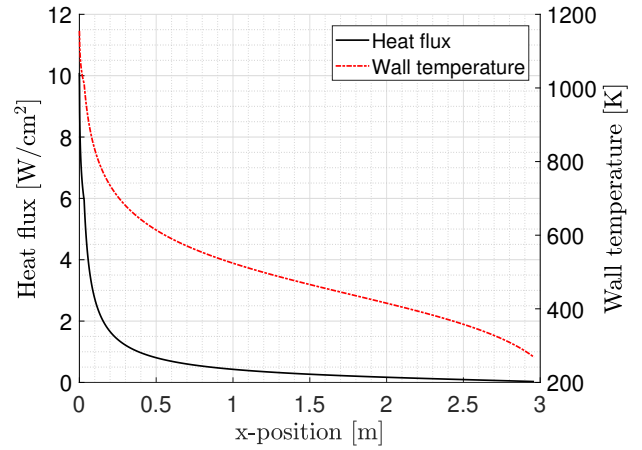


Figure 17: Stagnation temperatures during ascent

A number of options were considered to try and further reduce the peak temperatures experienced during ascent. In conjunction with [7] the initial ascent trajectory was changed such that the rocket was travelling slower in the lower atmosphere, where higher densities lead to higher heating. This reduced the expected stagnation temperatures to 1200 K. Secondly an investigation into the variation of peak stagnation temperatures with radius of curvature was carried out. The radius of curvature of the Haack series nose is found to be 0.11 m. To see what effect the radius of curvature had on heating this was varied from 0.1 m to 0.4 m as shown in Figure 18.

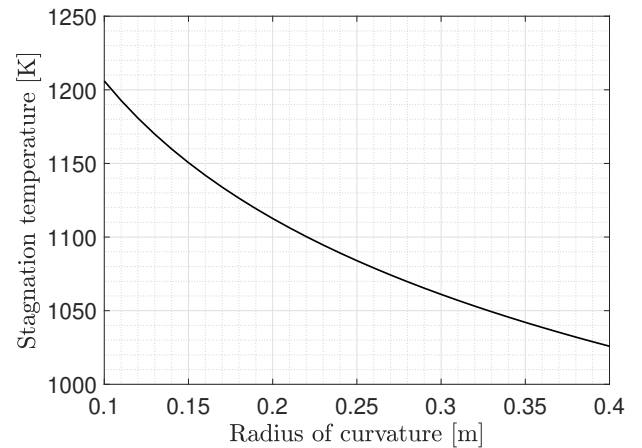


Figure 18: Variation in stagnation temperatures with radius of curvature

As can be seen, even by doubling the radius of curvature the stagnation temperature is only found to decrease by less than 100 K. However the pressure drag experienced by the rocket is also highly dependent on the radius of curvature, and

a two fold increase in radius of curvature is found to almost double the drag as discussed further in [3]. For this reason it was decided to keep the nose cone as an unmodified Haack series geometry.

7 Second Stage Recovery Feasibility

Although a business decision was made to not recover the second stage, a feasibility study was carried out to examine the possibility of second stage recovery for future missions. Since the second stage was not planned to be recovered, no detailed analysis was carried out for its EDL trajectory upon releasing its payload.

As such, a key assumption made in the subsequent analysis is that the re-entry conditions are the same as that of the space shuttle: 8000 m/s at 100 km (the Karman Line)[34]. Unlike for the first stage, an estimation for the C_d is not available, however for hypersonic re-entry speeds it is often assumed constant at 1.6 [6]. The mass of the empty second stage was found to be 159 kg, making the BC 87.8 kg/m².

if any radiative contributions. Since the second stage would have such a low BC, it would slow down quickly in the upper atmosphere where the density is very low. By the time it reaches a point where the density is greater, it will have slowed down enough such that radiative heating is very unlikely. In this case, LI-900 is not a suitable choice for the heat shield since the heat flux is at the limit of that sustainable for a thermal sink. Using PICA as the second stage heat shield material, the thickness distribution is given by Figure 19.

Interestingly, the thickness of the shield for the second stage is actually less than that of the first stage, despite having almost double the heat load. This comes from the engineering estimations in for heat shield thickness which is found to be inversely proportional to entry velocity, using a different material, and as a result of having such a low BC. The mass of the required heat shield was found to be 24 kg forming 15 % of the total mass of the second stage compared to the first stage heat shield which was just 3 % of the first stage weight. Combined with the extra costs associated with recovery of the second stage, it makes it unlikely that recovery of the second stage is a viable option.

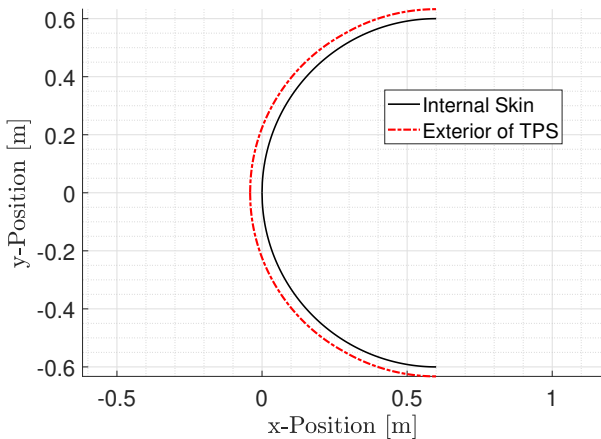


Figure 19: Variation in PICA thickness for second stage EDL

The maximum heat flux the second stage would be subjected to was 97.2 W/cm². This is still a relatively small heat flux and is due to the extremely low BC of the second stage. The heat load for which the second stage should withstand was found to be 4.3 kJ/cm². Once again heat flux due to radiative heat transfer is neglected, as even for speeds of 8000 m/s, Figure 6 shows little

8 Discussion

To try and validate the results, or at least have a sense of the numbers, comparison was made throughout to current and past re-entry spacecraft. To a first order the temperatures and heat flux calculated in this report are very similar to that experienced in the space shuttle (50 W/cm², 1500 K [16]). Further analysis is carried out in [4] with the use of CFD. It was found that CFD predicted temperatures roughly 200 K hotter than initially predicted, and is attributed to the approximate nature of the Centreline Heating method and not accounting for real gas effects in the CFD simulations due to convergence issues. Overall this confirms that the analysis is valid, at least for a first iteration.

During the first stage analysis it was assumed that the EDL body was hemispherical. Once the TPS had been designed, the thickness distribution changed the effective shape of the heat shield

such that it was no longer hemispherical. This may change the heat flux distribution and so the heating should be re analysed for the new shape. This would be an iterative procedure and should be carried out until there is little variation in TPS geometry.

This analysis only considers the contributions from convective heat transfer as radiative heat transfer was shown to be minimal. However, there are other heating mechanisms which have not been accounted for. Entropy swallowing is one such phenomena in which entropy gradients behind a curved bow shock can increase heating by the order of 50 % [16] and is discussed further in [3]. Ultimately, it was concluded that a more sophisticated analysis would be required to predict entropy effects.

It should be noted that in order for this to occur, the boundary layer must grow such that it entraps streamlines which have crossed the shock at different points. Given that the heat shield is only of 0.6 m radius, and that transition was not predicted (thus the boundary layer will remain small) it is unlikely that the boundary layer will grow enough to swallow streamlines of different entropy in just 0.6 m, and entropy effects will be minimal. It may however contribute to heating along the body where the boundary layer has time to develop.

The transition Reynolds number was found using an approximate relationship, although the study was carried out on cones in hypersonic flows and is not directly applicable. Since turbulence can have a large impact on heat transfer more accurate predictions should be developed. Given the difficulty of predicting transition this may require wind tunnel testing. During analysis of the heat shield performance, a one dimensional heat transfer equation is used. In reality the body is three dimensional, and to more accurately predict the heat flux through the TPS, the heat transfer *along* the heat shield, as well as through the heat shield should be modelled.

Another area for improvement could be to try and account for surface roughness. A rougher surface will promote transition and lead to the higher heat flux associated with turbulent boundary

layers. Further more the radius of curvature was found to be a very important parameter in determining heat flux. Thus a flat plate with infinite radius of curvature would predict zero heat flux. Clearly this is not correct and instead an effective nose radius should be used, in which the streamlines and boundary layer form an “effective” geometry which should be analysed instead. An analytical method for estimating effective radius of curvature can be found in [11].

One last note to make is that the TPS sizing estimation used is only approximate. The equation to find the thickness of LI-900 models the surface as being uncoated, resulting in an over estimate of upto 18.7% in thickness due to uncoated surfaces having a lower surface emittance [26]. This would suggest the current design for the heat shield is over designed, and confirms why internal temperatures rise by just 50 K, when in reality they could afford to rise by a bit more and lose some extra weight associated with the heat shield.

9 Conclusions

Using a 3 DoF model the trajectory of the first stage EDL is calculated. Implementing the Centreline Heating methodology, the peak heat flux the first stage would experience on EDL was calculated to be 44 W/cm^2 . LI-900 was selected as the TPS material, varying in thickness from 18-0.7 cm for a total weight of 30 kg. Subsequent analysis showed that the TPS was sufficient to protect the spacecraft, with internal temperatures rising by just over 50 K. Some limitations were noted in the accuracy of the heating predictions and analysis but by comparison with existing data and CFD was found to be first order accurate. An investigation into the ascent trajectory found peak temperatures of 1200 K, such that heating would not be an issue on ascent as it is below the operational temperature of the ceramic used for the nose fairing. Further more a feasibility study into bringing back the second stage concluded that the heat shield would form 15 % of the empty weight, making recovery uneconomical. Improvements to predictions can be made by use of specialised heating software such as CBAERO [35] or NASA’s FIAT [36] software.

References

- [1] Access to space. <http://ec.europa.eu/research/participants/portal/desktop/en/opportunities/h2020/topics/space-16-tec-2018.html>. [Accessed: 2018-06-15].
- [2] Trieu Ho. *A management approach to the design of a small commercial launcher*. Imperial College London, 2018.
- [3] Nicolas Gomez. *Aerothermodynamics of a Low Earth Orbiter Launcher: Hypersonic Methods and Analysis*. Imperial College London, 2018.
- [4] Alexander Kent. *Aerothermodynamics of a Low Earth Orbiter Launcher: Subsonic and Heating Analysis and Validation*. Imperial College London, 2018.
- [5] Giulio Avanzini. *Entry, Descent, Landing and Ascent*. Politecnico di Torino, 2009.
- [6] Max Braun, Paul Bruce, and Errikos Levis. Strategies to utilize advanced heat shield technology for high-payload mars atmospheric entry missions. *Acta Astronautica*, 136:22–33, 2017.
- [7] Lachlan Orr. *Systems Intergration, Mission Planning and Control*. Imperial College London, 2018.
- [8] Standard Atmosphere. Standard, International Organization for Standardization, 1975. ISO 2533:1975/Add21997(E/F/R).
- [9] Report 1135: Equations, Tables and Charts for Compressible Flow. Technical report, National Advisory Committee for Aeronautics, 1953.
- [10] John David Anderson Jr. *Fundamentals of aerodynamics*. Tata McGraw-Hill Education, 2010.
- [11] Scott K Martinelli and Robert D Braun. Centerline heating methodology for use in preliminary design studies. In *Aerospace Conference, 2011 IEEE*, pages 1–18. IEEE, 2011.
- [12] Kenneth Sutton and Randolph A Graves Jr. *A general stagnation-point convective heating equation for arbitrary gas mixtures*. 1971.
- [13] H Julian Allen and Alfred J Eggers Jr. A study of the motion and aerodynamic heating of missiles entering the earth’s atmosphere at high supersonic speeds. 1953.
- [14] James A Fay. Theory of stagnation point heat transfer in dissociated air. *Journal of the Aerospace Sciences*, 25(2):73–85, 1958.
- [15] John J Bertin. *Hypersonic aerothermodynamics*. AIAA, 1994.
- [16] JD Anderson Jr. *Hypersonic and High Temperature Gas Dynamics*. AIAA Publications, AIAA, Reston, VA, 2000.
- [17] JN Moss EV Zoby and K Sutton. Approximate convective-heating equations for hypersonic flows. *Journal of Spacecraft and Rockets*, 18(1):64–70, 1981.
- [18] Ernst R Eckert. Survey on heat transfer at high speeds. Technical report, MINNESOTA UNIV MINNEAPOLIS, 1961.
- [19] Diego Capriotti. Viscous optimized hypersonic waveriders. In *25th AIAA Aerospace Sciences Meeting*, page 272, 1987.
- [20] Michael E Tauber and Kenneth Sutton. Stagnation-point radiative heating relations for earth and mars entries. *Journal of Spacecraft and Rockets*, 28(1):40–42, 1991.

- [21] David Akin. Applications of ultra-low ballistic coefficient entry vehicles to existing and future space missions. In *SpaceOps 2010 Conference Delivering on the Dream Hosted by NASA Marshall Space Flight Center and Organized by AIAA*, page 1928, 2010.
- [22] E Fantino, M Grassi, P Pasolini, F Causa, C Molfese, R Aurigemma, N Cimminiello, D de la Torre, P Dell’Aversana, F Esposito, et al. The small mars system. *Acta Astronautica*, 137:168–181, 2017.
- [23] E Bassano, R Savino, R Lo Forti, A Ferrarotti, C Richiello, G Russo, R Aurigemma, F Punzo, and P Dell’Aversana. Irene–italian re-entry nacelle for microgravity experiments. In *62nd International Astronautical Congress, Cape Town (South Africa)*, 2011.
- [24] Nasa - irve-3: Inflatable heat shield a splashing success. https://www.nasa.gov/directorates/spacetech/game_changing_development/HIAD/irve3-success.html. [Accessed: 2018-06-11].
- [25] Valerio Carandente, Roberto Scigliano, Valeria De Simone, Antonio Del Vecchio, and Roberto Gardi. A finite element approach for the design of innovative ablative materials for space applications. In *6th European conference for aeronautics and space sciences (EUCASS)*. Krakow, Poland, 2015.
- [26] Steven Sepka and Jamshid A Samareh. Thermal protection system mass estimating relationships for blunt-body, earth entry spacecraft. In *45th AIAA Thermophysics Conference*, page 2507, 2015.
- [27] Thermal protection systems. https://www.nasa.gov/centers/johnson/pdf/584728main_Wings-ch4b-pgs182-199.pdf. [Accessed: 2018-06-11].
- [28] Pica heat shield. <http://www.spacex.com/news/2013/04/04/pica-heat-shield>. [Accessed: 2018-06-11].
- [29] John Dec and Robert Braun. An approximate ablative thermal protection system sizing tool for entry system design. In *44th AIAA Aerospace Sciences Meeting and Exhibit*, page 780, 2006.
- [30] Nasa offers space shuttle tiles to schools. <https://www.space.com/9641-nasa-offers-space-shuttle-tiles-schools.html>. [Accessed: 2018-06-13].
- [31] WY Walter, T Ezra, and C George. Combined conductive/radiative heat transfer in high porosity fibrous insulation materials: theory and experiment. In *The 6th ASME-JSME thermal engineering joint conference*, 2003.
- [32] Patrick Hanlon. *Rocket Structural Design: Material Selection and structural layout*. Imperial College London, 2018.
- [33] Granta. CES EduPack 2017, 2017. Computer Software.
- [34] Returning from space: Re-entry. https://www.faa.gov/about/office_org/headquarters_offices/avs/offices/aam/cami/library/online_libraries/aerospace_medicine/tutorial/media/III.4.1.7_Returning_from_Space.pdf. [Accessed: 2018-06-14].
- [35] David Kinney. Aero-thermodynamics for conceptual design. In *42nd AIAA Aerospace Sciences Meeting and Exhibit*, page 31, 2004.
- [36] YK Chen and FS Milos. Fully implicit ablation and thermal analysis program (FIAT). *Journal of Spacecraft and Rockets*, 36(3):475–483, 1999.

Appendix A: Body geometry

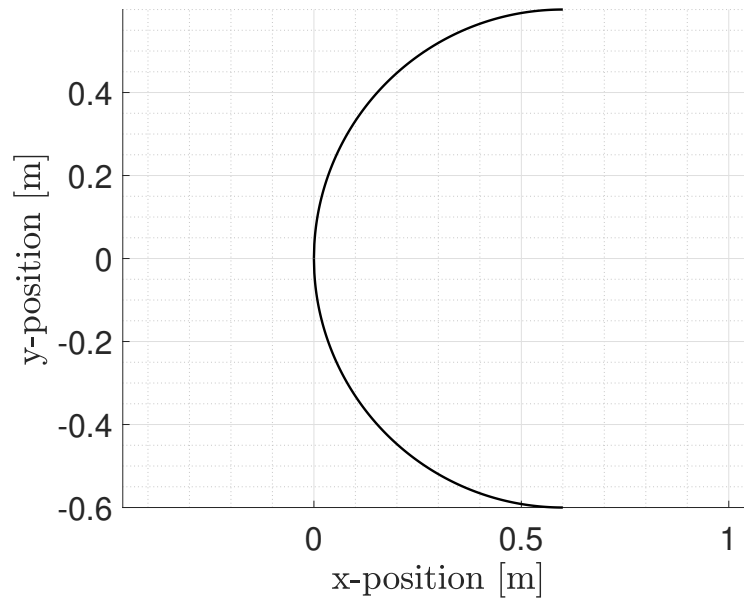


Figure 20: Blunt body re-entry geometry

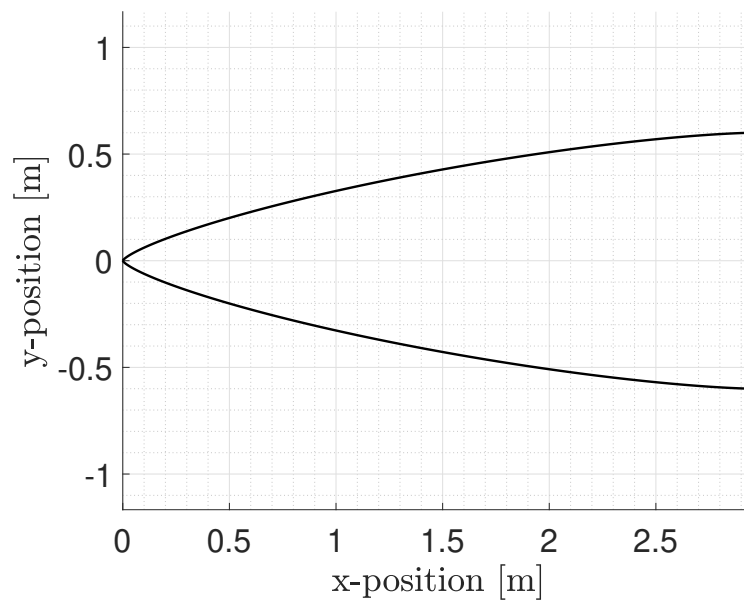


Figure 21: Haack series nose cone geometry on ascent

Appendix B: Heating Analysis of first stage EDL

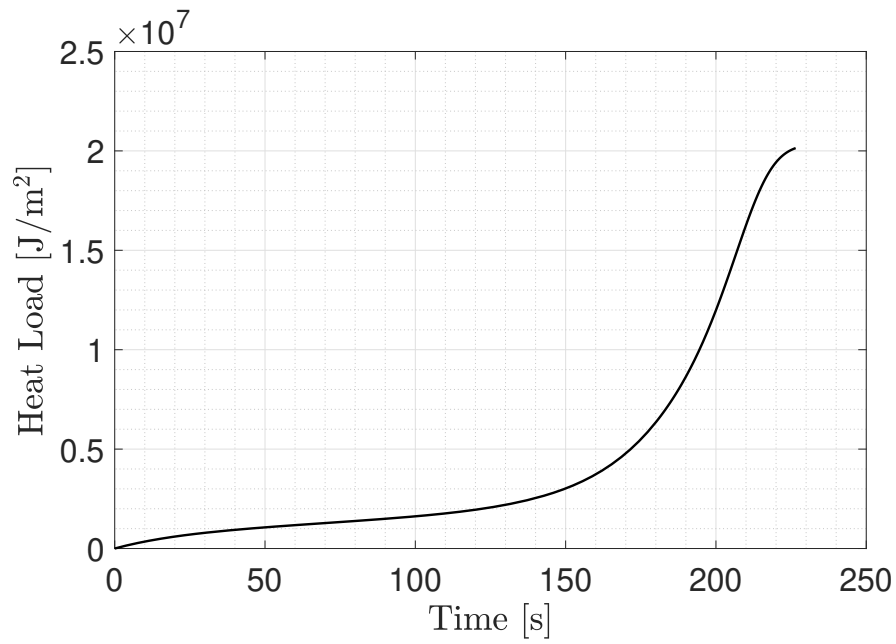


Figure 22: Change in heat load during first stage EDL

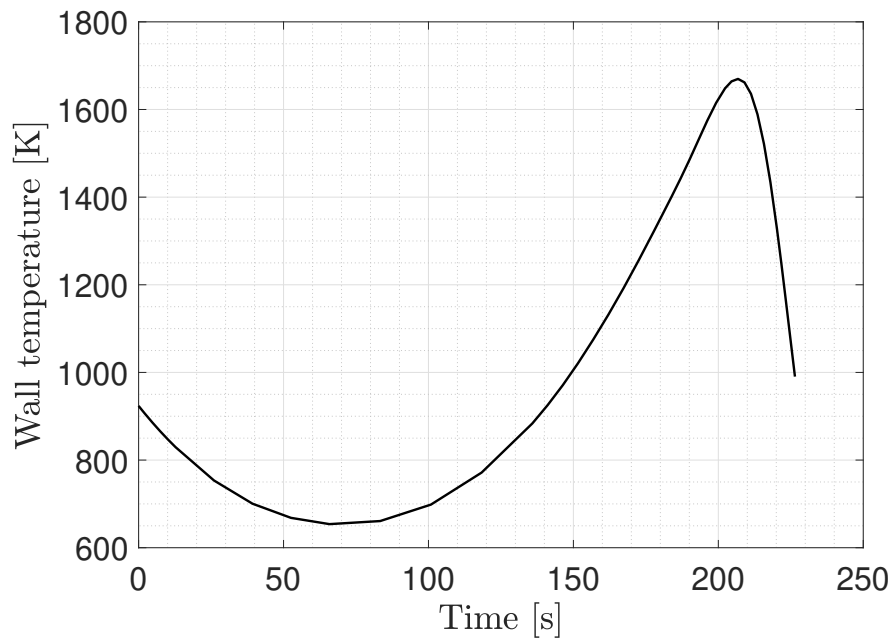


Figure 23: Variation in stagnation temperatures during first stage EDL